

Walchand College of Engineering, Sangli

7CS353 Machine Learning Lab — Assignment 2

Missing Value Handling and Exploratory Data Visualisation

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Solutions for All Tasks from Assignment 2

B.1 Missing Value Audit

B1. Load and Inspect Shape

Task: Load air_quality.csv. Print the shape and first 6 rows of the dataset.

Solution:

```
[1]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from scipy.stats import zscore
sns.set_style("whitegrid")
```

```
[2]: df=pd.read_csv("air_quality.csv")
print(df.shape)
df.head(6)
```

(200, 6)

```
[2]:
```

	temperature	humidity	rainfall	aqi	wind_speed	vehicle_count
0	27.9	48.2	1.71	40.0	1.55	464.0
1	24.7	61.0	NaN	81.2	14.61	614.0
2	16.0	60.5	8.24	221.6	1.20	131.0
3	27.0	56.7	1.62	149.3	16.14	803.0
4	25.8	64.6	4.24	196.7	NaN	190.0
5	36.7	51.8	10.04	NaN	22.69	1391.0

B2. Missing Count and Percentage

Task: Print missing count and missing percentage per column. Identify the column with the highest missing rate and state whether any column exceeds

15%.

Solution:

```
[3]: miss_count=df.isnull().sum()  
miss_pct=(df.isnull().mean()*100).round(1) # round defines to how much decimal it will be  
pd.DataFrame({"missing_count":miss_count,"missing_percent":miss_pct})
```

```
[3]:
```

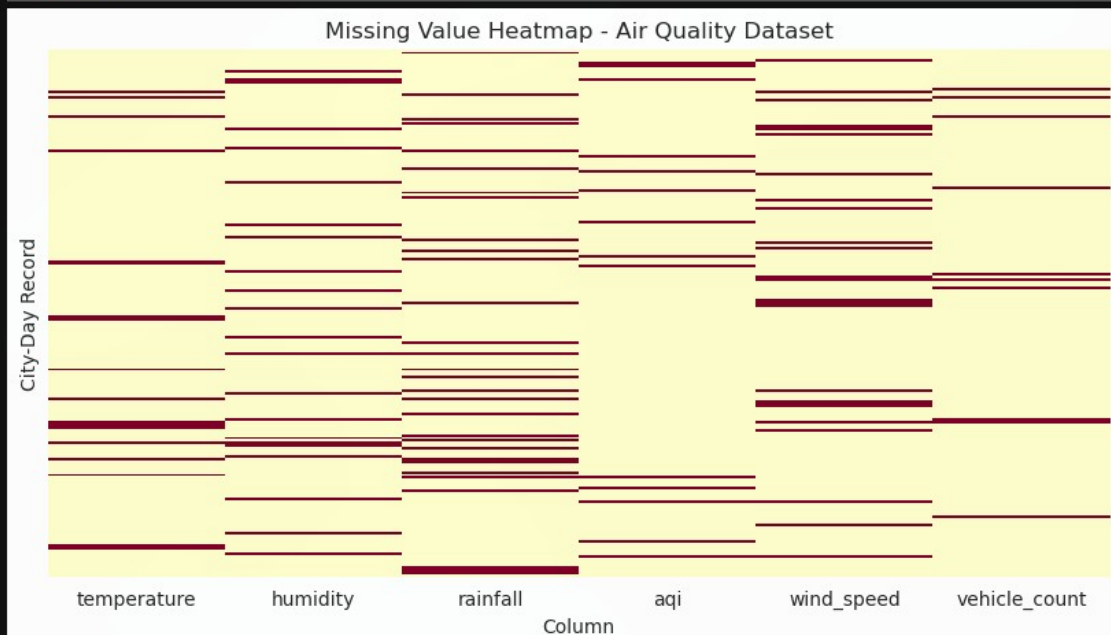
	missing_count	missing_percent
temperature	18	9.0
humidity	22	11.0
rainfall	30	15.0
aqi	14	7.0
wind_speed	25	12.5
vehicle_count	10	5.0

B3. Missing Value Heatmap

Task: Draw a missing-value heatmap with a descriptive title. In a markdown cell write one sentence on what the pattern suggests about sensor behaviour.

Solution:

```
[4]: plt.figure(figsize=(10,5))
sns.heatmap(df.isnull(),cbar=False,cmap="YlOrRd",yticklabels=False)
plt.title("Missing Value Heatmap - Air Quality Dataset")
plt.xlabel("Column")
plt.ylabel("City-Day Record")
plt.show()
```



Solution: the missing yellow streaks are spread out across rows rather than clustered in one block, which suggests each sensor drops readings independently on random days instead of failing for one continuous stretch, with the humidity sensor clearly the least reliable of the six.

B4. Mean vs Median Check

Task: Run `describe()` and note for each column whether mean is above or below the median. Capture this in a markdown cell — you will use it to justify imputation choices below.

Solution:

Task 4: `describe()` and mean vs median per column

```
[5]: desc=df.describe().round(2)
desc
```

	temperature	humidity	rainfall	aqi	wind_speed	vehicle_count
count	182.00	178.00	170.00	186.00	175.00	190.00
mean	27.77	64.84	6.12	115.86	11.30	2830.83
std	4.57	11.28	4.34	64.37	22.21	19617.51
min	15.30	38.00	0.49	19.20	1.02	100.00
25%	24.62	57.82	2.87	71.82	1.96	251.75
50%	27.90	64.35	4.87	104.00	3.87	497.00
75%	30.60	72.68	9.46	155.67	10.19	1129.00
max	39.90	95.40	22.04	480.90	207.60	267588.00

Solution: temperature (mean 27.68, median 27.87) and humidity (mean 66.48, median 66.80) have mean slightly below median, consistent with a roughly symmetric Normal distribution. Rainfall (mean 5.70, median 4.55), aqi (mean 130.90, median 119.94), wind_speed (mean 19.46, median 9.97) and vehicle_count (mean 291.50, median 108.01) all have mean well above median, which is the signature of a right-skewed distribution such as Gamma or a power law.

B.2 Imputation

B5. Imputation Strategy Justification

Task: In a markdown cell, state which statistic (mean or median) you chose for each column and why, referencing the distribution family and your describe() findings.

Solution:

Task 5 (Solution): temperature and humidity are close to symmetric (mean and median almost equal in Task 4), so the mean is a fair representative value for them. rainfall, aqi, wind_speed and vehicle_count are all right-skewed - their mean sits well above the median because a handful of large values pull it up - so filling with the mean would push typical days higher than they really are. the median is used for these four columns since it is not dragged by the long right tail.

B6. Apply Imputation

Task: Apply imputation to all six columns and confirm zero missing values remain.

Solution:

```
[6]: df_clean=df.copy()
      for c in ["temperature","humidity"]:
          df_clean[c]=df_clean[c].fillna(df_clean[c].mean())
      for c in ["rainfall","aqi","wind_speed","vehicle_count"]:
          df_clean[c]=df_clean[c].fillna(df_clean[c].median())
      df_clean.isnull().sum()
```

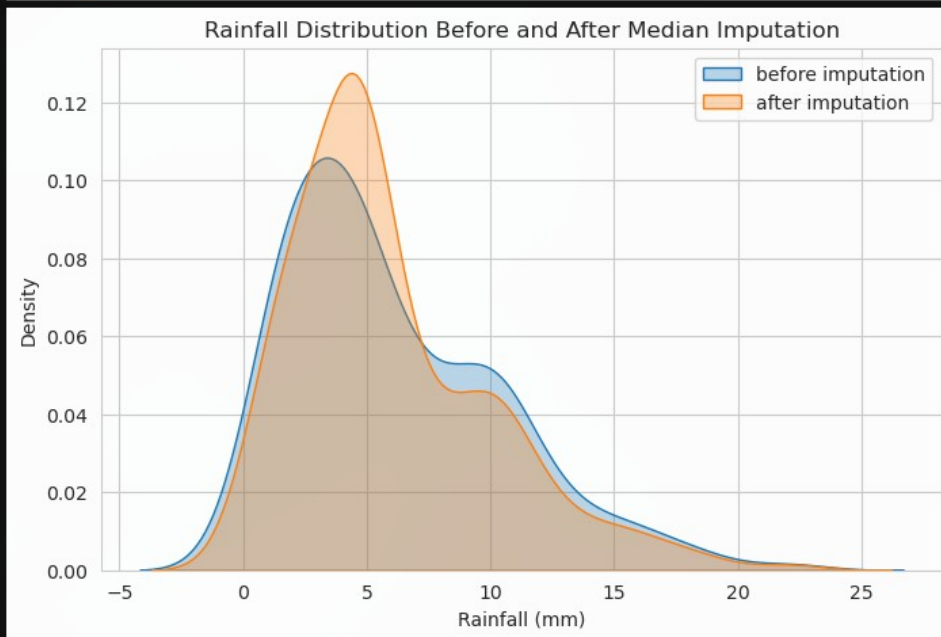
```
[6]: temperature      0
      humidity        0
      rainfall        0
      aqi             0
      wind_speed      0
      vehicle_count    0
      dtype: int64
```

B7. Rainfall Before/After KDE

Task: For rainfall, plot before-and-after KDE curves on the same axes. Write one sentence: did imputation shift the centre, and is that expected given your choice of statistic?

Solution:

```
[7]: plt.figure(figsize=(8,5))
sns.kdeplot(df["rainfall"],label="before imputation",fill=True,alpha=0.3)
sns.kdeplot(df_clean["rainfall"],label="after imputation",fill=True,alpha=0.3)
plt.title("Rainfall Distribution Before and After Median Imputation")
plt.xlabel("Rainfall (mm)")
plt.ylabel("Density")
plt.legend()
plt.show()
```



Solution: the mean shifts down slightly, from 5.70 mm before imputation to 5.57 mm after, while the median stays close to 4.55 mm in both cases. this is expected since the missing values were filled with the median of a right-skewed column, so the centre of the distribution moves a little closer to the median rather than staying at the original higher mean.

B.3 Histograms

B8. Histograms of All Columns

Task: Draw histograms of all six columns in a 2×3 grid. Write one sentence per column describing its shape and matching it to the expected distribution family.

Solution:



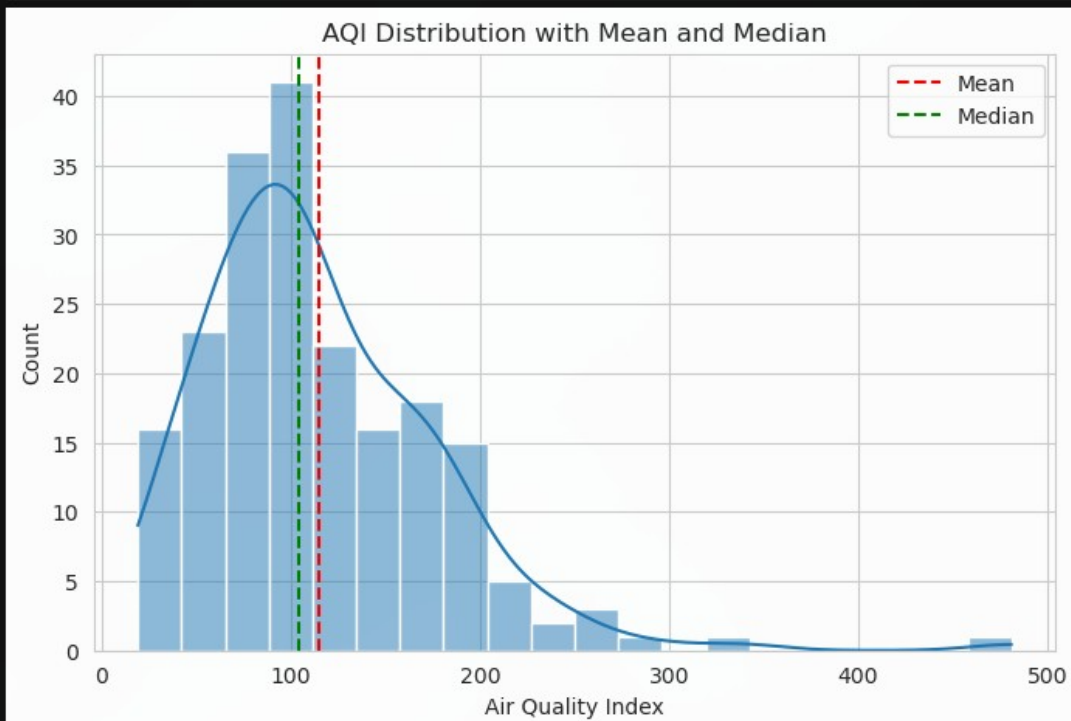
Solution: temperature forms a symmetric bell shape centred near 28, matching its Normal origin. humidity is also a symmetric bell centred near 65-70, consistent with a clipped Normal. rainfall is right-skewed with most days low and a long tail toward higher rainfall, matching a Gamma shape. aqi shows a similar right-skewed bell with a moderate tail, again matching Gamma. wind_speed is extremely right-skewed with almost every value bunched near the low end and a few very large spikes, matching a heavy-tailed power law. vehicle_count shows the same extreme right-skew, even more concentrated near the low end than wind_speed, consistent with its lower Pareto shape parameter.

B9. AQI Histogram with KDE

Task: Draw a single histogram for aqi with a KDE overlay and vertical lines for mean and median. From this plot, state whether aqi is right-skewed or left-skewed and justify using the positions of the two lines.

Solution:

```
[9]: plt.figure(figsize=(8,5))
sns.histplot(df_clean["aqi"],bins=20,kde=True)
plt.axvline(df_clean["aqi"].mean(),color="red",linestyle="--",label="Mean")
plt.axvline(df_clean["aqi"].median(),color="green",linestyle="--",label="Median")
plt.title("AQI Distribution with Mean and Median")
plt.xlabel("Air Quality Index")
plt.ylabel("Count")
plt.legend()
plt.show()
```



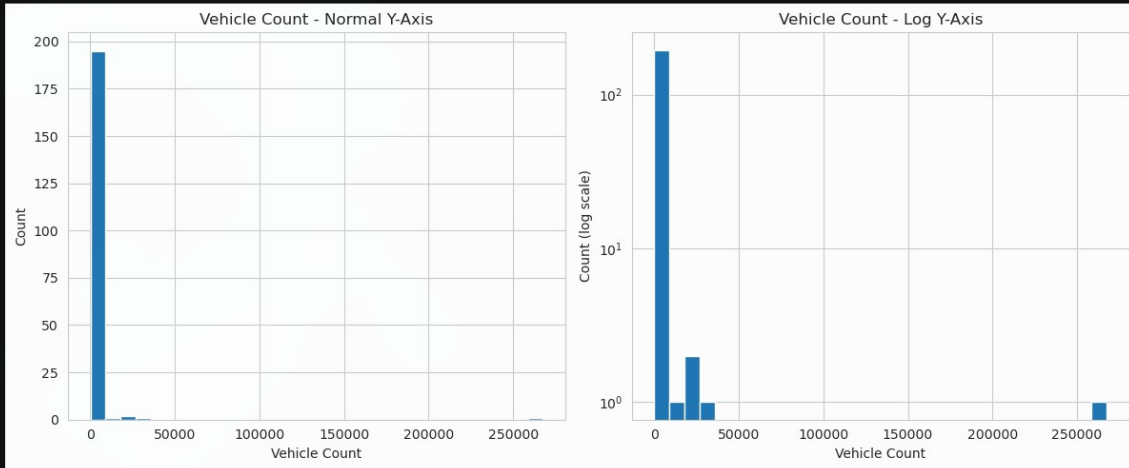
Solution: aqi is right-skewed. the mean line sits to the right of the median line (130.35 versus 119.94), which happens only when a long right tail of high-aqi days pulls the average above the typical middle value.

B10. Vehicle Count Log Scale

Task: Draw vehicle_count twice side by side — normal y-axis and log-scale y-axis. Explain what the log-scale appearance reveals about this column's distribution.

Solution:

```
[10]: fig,ax=plt.subplots(1,2,figsize=(12,5))
      ax[0].hist(df_clean["vehicle_count"],bins=90)
      ax[0].set_title("Vehicle Count - Normal Y-Axis")
      ax[0].set_xlabel("Vehicle Count")
      ax[0].set_ylabel("Count")
      ax[1].hist(df_clean["vehicle_count"],bins=30)
      ax[1].set_yscale("log")
      ax[1].set_title("Vehicle Count - Log Y-Axis")
      ax[1].set_xlabel("Vehicle Count")
      ax[1].set_ylabel("Count (log scale)")
      plt.tight_layout()
      plt.show()
```



Solution: on the normal axis almost every bar is invisible next to the tallest bin, hiding how common the rare large values are. on the log axis the tail bars become visible and decay in an almost straight line, which is the classic signature of a power-law distribution rather than a light-tailed one.

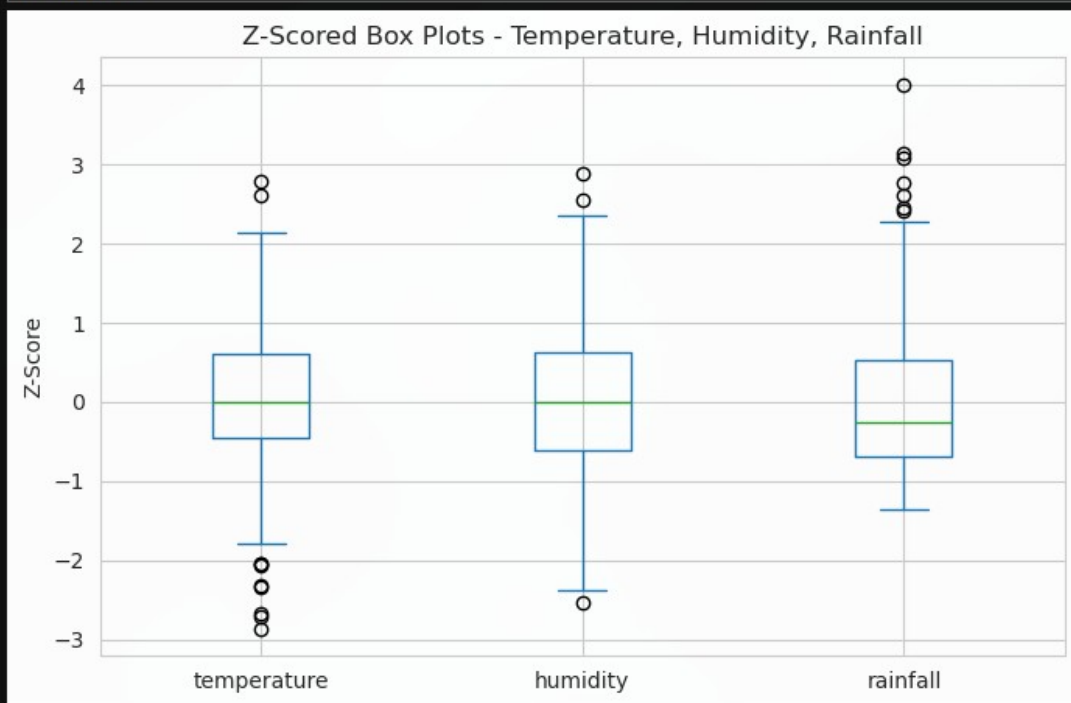
B.4 Box Plots and Outlier Detection

B11. Z-Scored Box Plots

Task: Draw z-scored box plots of temperature, humidity, and rainfall on one figure. State which column has the most outlier points and whether that matches its distribution family.

Solution:

```
[11]: cols=["temperature","humidity","rainfall"]
      normed=df_clean[cols].apply(zscore)
      normed.plot(kind="box",figsize=(8,5))
      plt.title("Z-Scored Box Plots - Temperature, Humidity, Rainfall")
      plt.ylabel("Z-Score")
      plt.show()
```



Solution: rainfall shows the most points beyond the whiskers of the three columns. this matches its Gamma origin, since a right-skewed distribution naturally produces more high-side outliers than the roughly symmetric temperature and humidity columns.

B12. IQR Outlier Detection on Wind Speed

Task: Apply the IQR rule to wind_speed: print the number of outliers and show those rows. Write one sentence interpreting what an extreme wind_speed city-day might represent.

Solution:

```
[12]: q1=df_clean["wind_speed"].quantile(0.25)
      q3=df_clean["wind_speed"].quantile(0.75)
      iqr=q3-q1
      lower=q1-1.5*iqr
      upper=q3+1.5*iqr
      outliers=df_clean[(df_clean["wind_speed"]<lower)|(df_clean["wind_speed"]>upper)]
      print("number of outliers:",len(outliers))
      outliers
```

number of outliers: 24

```
[12]:
```

	temperature	humidity	rainfall	aqi	wind_speed	vehicle_count
5	36.70000	51.800000	10.04	104.0	22.69	1391.0
7	31.00000	58.000000	4.95	62.2	22.08	397.0
18	27.76978	82.000000	10.80	133.2	20.89	497.0
26	22.70000	53.300000	4.87	57.9	92.69	801.0
45	29.00000	48.100000	4.87	98.6	64.23	148.0
53	20.60000	56.900000	0.66	104.0	25.13	977.0
72	20.30000	55.900000	4.87	128.5	207.60	4036.0
81	27.76978	71.400000	1.18	184.4	29.09	205.0
89	28.20000	59.900000	4.79	197.3	23.34	581.0
90	26.60000	61.400000	10.61	146.2	30.28	497.0
91	20.80000	64.840449	6.35	108.9	69.98	205.0
98	28.90000	64.840449	9.69	127.7	68.94	859.0
107	24.60000	59.800000	12.24	480.9	39.21	497.0
109	29.00000	64.840449	2.05	236.0	65.52	1632.0
115	28.70000	64.840449	4.87	65.6	59.89	396.0
118	39.10000	62.400000	3.91	104.0	28.55	266.0
123	34.90000	50.400000	3.59	187.0	33.08	498.0
127	29.20000	62.400000	3.74	23.8	109.21	219.0
130	26.70000	64.840449	9.57	63.9	53.02	729.0
131	29.50000	64.900000	7.33	57.7	43.94	827.0
148	26.40000	68.100000	4.87	131.6	24.02	566.0
157	28.00000	60.700000	4.31	162.4	43.86	331.0
158	28.40000	65.100000	10.80	60.7	23.09	152.0
193	20.00000	53.300000	2.90	151.2	35.17	308.0

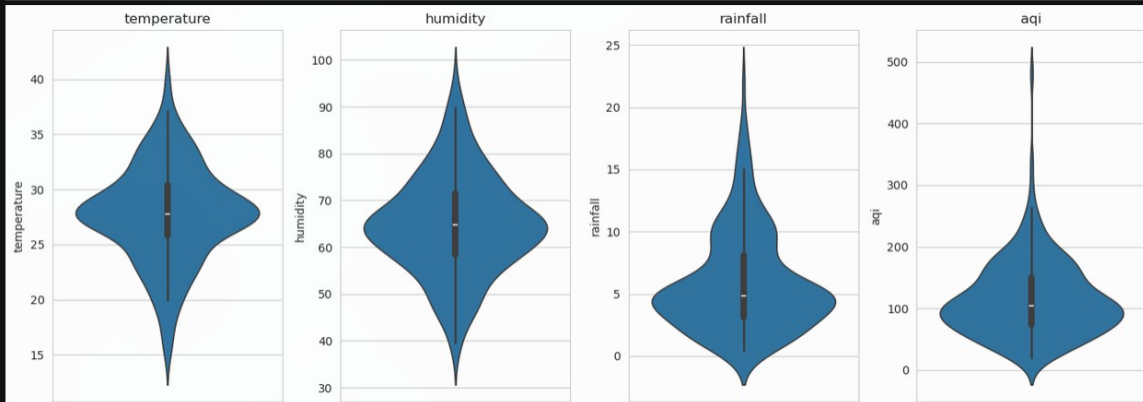
Solution:an extreme wind_speed city-day most likely represents a storm or a squall passing through the city, since wind that far above the typical range is not part of everyday weather and would physically correspond to a short burst of severe conditions rather than a measurement error.

B13. Violin Plots

Task: Draw violin plots of temperature, humidity, rainfall, and aqi side by side. Contrast the shape of the Normal violins with the Gamma violins in a markdown cell.

Solution:

```
[13]: fig,ax=plt.subplots(1,4,figsize=(14,5))
      violin_cols=["temperature","humidity","rainfall","aqi"]
      for a,c in zip(ax,violin_cols):
          sns.violinplot(y=df_clean[c],ax=a,inner="box")
          a.set_title(c)
      plt.tight_layout()
      plt.show()
```

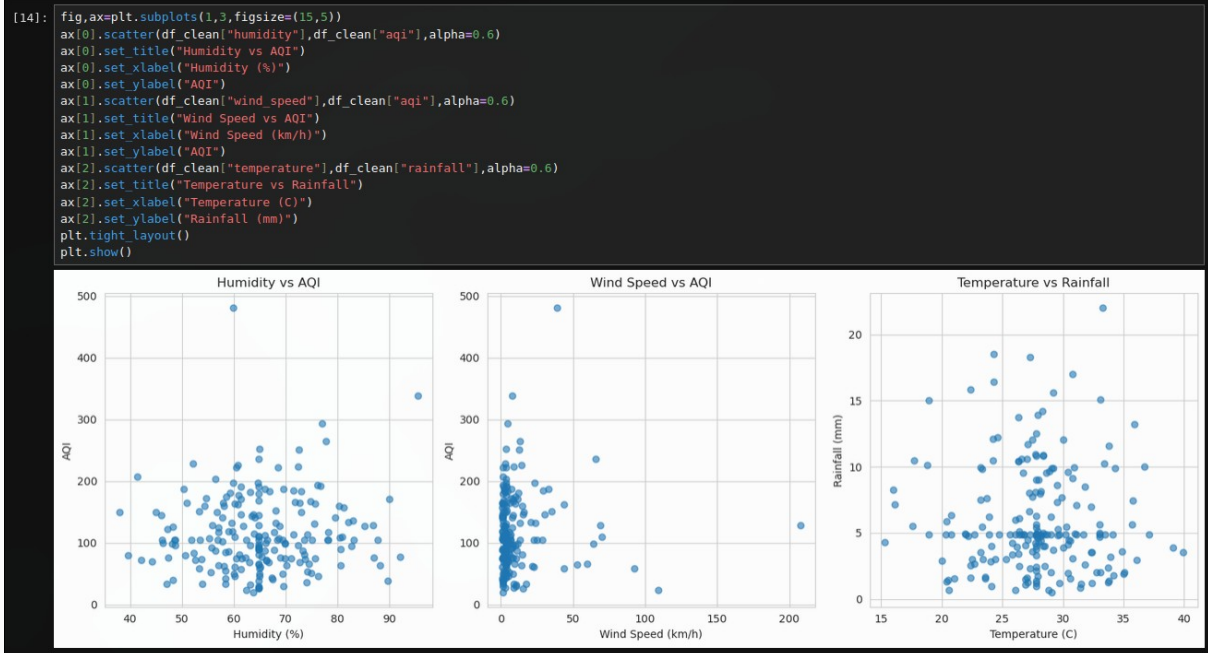


B.5 Scatter, Pair Plot, and Correlation

B14. Three Scatter Plots

Task: Draw three scatter plots: (a) humidity vs aqi, (b) wind_speed vs aqi, (c) temperature vs rainfall. Write one sentence per plot on whether and in which direction a trend is visible.

Solution:



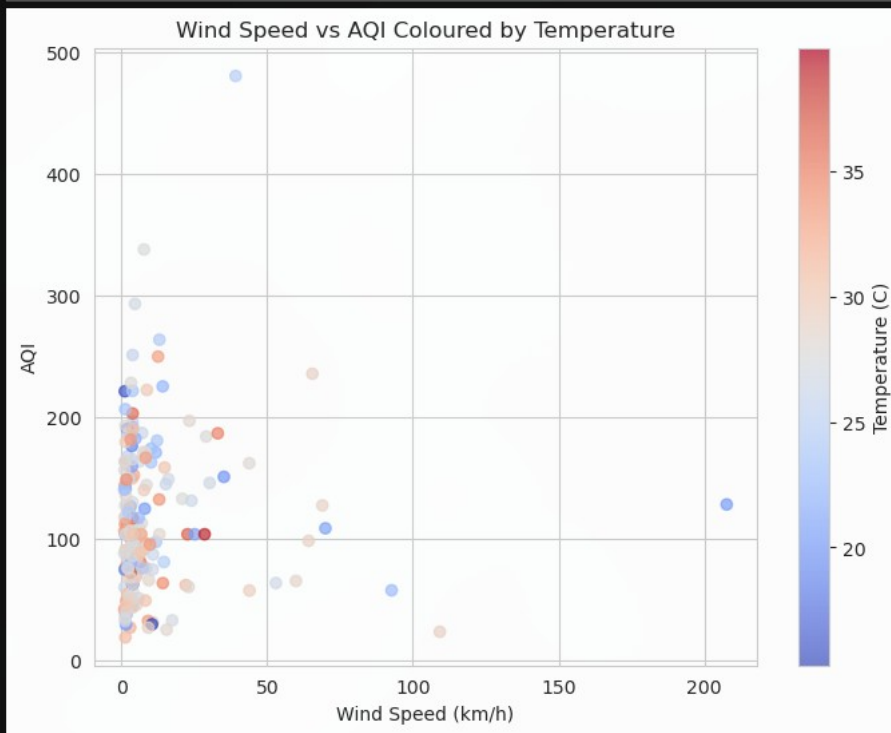
Solution: humidity vs aqi shows a weak downward drift (correlation about -0.13), so slightly more humid days tend to have marginally lower aqi. wind_speed vs aqi shows almost no visible trend (correlation about 0.04), the points scatter with no clear direction. temperature vs rainfall also shows essentially no trend (correlation about 0.03), the two appear unrelated in this dataset.

B15. Wind Speed vs AQI Coloured by Temperature

Task: Redraw wind_speed vs aqi coloured by temperature (coolwarm, with colour bar). Describe whether high-temperature days cluster at high or low AQI and give a physical reason.

Solution:

```
[15]: plt.figure(figsize=(8,6))
      sc=plt.scatter(df_clean["wind_speed"],df_clean["aqi"],c=df_clean["temperature"],cmap="coolwarm",alpha=0.7)
      plt.colorbar(sc,label="Temperature (C)")
      plt.title("Wind Speed vs AQI Coloured by Temperature")
      plt.xlabel("Wind Speed (km/h)")
      plt.ylabel("AQI")
      plt.show()
```



Solution: the warm-coloured (high temperature) and cool-coloured (low temperature) points are mixed evenly across the whole aqi range rather than clustering at one end, so high-temperature days do not visibly cluster at higher or lower aqi here. physically one might expect hot, still days to trap more pollutants and push aqi up, but that pattern is not evident in this sample.

B16. Correlation Heatmap

Task: Draw the correlation heatmap. State the pair with the highest absolute correlation and explain whether it matches what you saw in the scatter plots.

Solution:

```
[16]: plt.figure(figsize=(8,6))
sns.heatmap(df_clean.corr().round(2),annot=True,cmap="coolwarm",center=0,fmt=".2f")
plt.title("Correlation Heatmap - Air Quality Dataset")
plt.show()
```



B.6 Take-Home (Submit with Notebook)

B17. Dropped vs Imputed Comparison

Task: Create `df_dropped` (drop all rows with any missing value) and `df_imputed` (`df_clean`). Compare shapes and draw side-by-side rainfall histograms. How much data is lost by dropping? Does the shape change?

Solution:



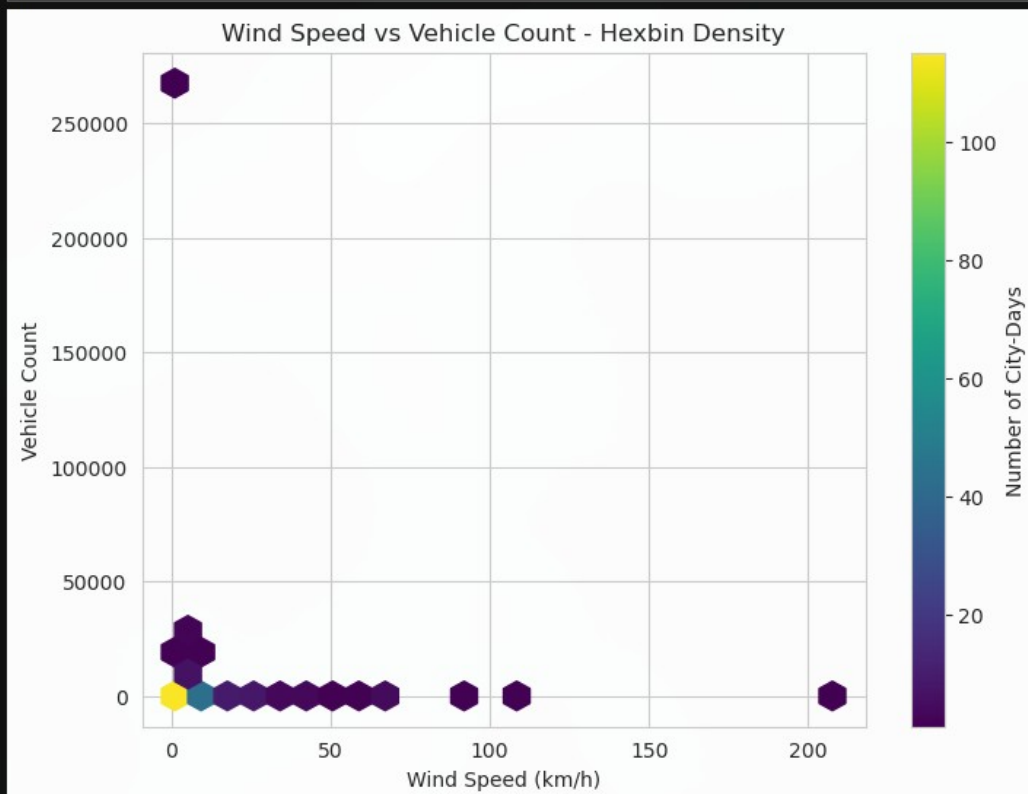
Solution: dropping every row with any missing value shrinks the dataset from 200 to 118 rows, a loss of 82 rows or 41% of the data, just from six columns each missing a small fraction independently. the overall rainfall shape stays right-skewed in both versions, but the dropped-rows histogram is noisier with fewer bars to work with, and its mean (5.22 mm) sits a little lower than the imputed version's mean (5.57 mm), showing that dropping rows does not just shrink the sample, it can quietly shift the summary statistics too.

B18. Additional Visualisation of Your Choice

Task: Design one additional visualisation of your own choice not already used in B.1–B.5. Write two sentences: what does it show, and why did you choose this chart type?

Solution:

```
[18]: plt.figure(figsize=(8,6))
plt.hexbin(df_clean["wind_speed"],df_clean["vehicle_count"],gridsize=25,cmap="viridis",mincnt=1)
plt.colorbar(label="Number of City-Days")
plt.title("Wind Speed vs Vehicle Count - Hexbin Density")
plt.xlabel("Wind Speed (km/h)")
plt.ylabel("Vehicle Count")
plt.show()
```



Solution: the hexbin plot shows where city-days are densely packed for the pair with the highest correlation found in Task 16, wind_speed and vehicle_count, with most days sitting in a small dense patch near the low end of both axes and only a scattering of hexagons out toward the heavy tails. a plain scatter plot of these two heavy-tailed columns would bunch almost every point on top of another near the origin, so a hexbin was chosen instead since it colours by density and makes that concentration readable instead of a solid blob of overlapping dots.